STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-ACYCLIC-GOOD

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3444-3461 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 3463-3498, the complete author proof (34 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 18 of queue_sdga.json (pb=1442, stmt 3444-3461, proof 3463-3498).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0FSH: sources/stacks-sdga/pinned-original-source.tex, source lines 3186–3196. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{A}$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. Let $\mathcal{P}$ be a good acyclic right differential graded $\mathcal{A}$-module.*

1. *for any differential graded left $\mathcal{A}$-module $\mathcal{N}$ the tensor product $\mathcal{P} \otimes_{\mathcal{A}} \mathcal{N}$ is acyclic,*
2. *for any morphism $(f, f^\sharp) : (Sh(\mathcal{C}'), \mathcal{O}') \to (Sh(\mathcal{C}), \mathcal{O})$ of ringed topoi and any differential graded $\mathcal{O}'$-algebra $\mathcal{A}'$ and any map $\varphi : f^{-1}\mathcal{A} \to \mathcal{A}'$ of differential graded $f^{-1}\mathcal{O}$-algebras the pullback $f^*\mathcal{P}$ is acyclic and good.*

Proof. Proof of (1). By Lemma 0FSH we can choose a good left differential graded $\mathcal{Q}$ and a quasi-isomorphism $\mathcal{Q} \to \mathcal{N}$. Then $\mathcal{P} \otimes_{\mathcal{A}} \mathcal{Q}$ is acyclic because $\mathcal{Q}$ is good. Let $\mathcal{N}'$ be the cone on the map $\mathcal{Q} \to \mathcal{N}$. Then $\mathcal{P} \otimes_{\mathcal{A}} \mathcal{N}'$ is acyclic because $\mathcal{P}$ is good and because $\mathcal{N}'$ is acyclic (as the cone on a quasi-isomorphism). We have a distinguished triangle

$$\mathcal{Q} \to \mathcal{N} \to \mathcal{N}' \to \mathcal{Q}[1]$$

in $K(Mod(\mathcal{A}, \mathrm{d}))$ by our construction of the triangulated structure. Since $\mathcal{P} \otimes_{\mathcal{A}} -$ sends distinguished triangles to distinguished triangles, we obtain a distinguished triangle

$$\mathcal{P} \otimes_{\mathcal{A}} \mathcal{Q} \to \mathcal{P} \otimes_{\mathcal{A}} \mathcal{N} \to \mathcal{P} \otimes_{\mathcal{A}} \mathcal{N}' \to \mathcal{P} \otimes_{\mathcal{A}} \mathcal{Q}[1]$$

in $K(Mod(\mathcal{O}))$. Thus we conclude.

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Proof of (2). Observe that $f^*\mathcal{P}$ is good by our definition of good modules. Recall that $f^*\mathcal{P} = f^{-1}\mathcal{P} \otimes_{f^{-1}\mathcal{A}} \mathcal{A}'$. Then $f^{-1}\mathcal{P}$ is a good acyclic (because f^{-1} is exact) differential graded $f^{-1}\mathcal{A}$-module. Hence we see that $f^*\mathcal{P}$ is acyclic by part (1). $\square$